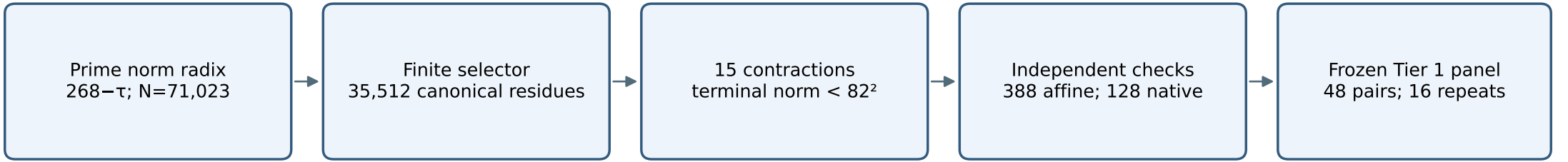
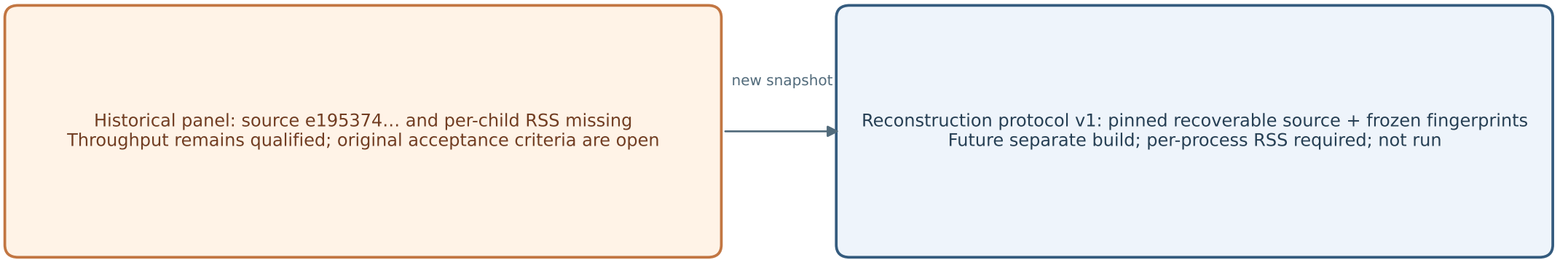


Proof scope: integer recoding and runtime tie rule; candidate-specific secp256k1 group bridge remains open

The throughput result covers repeated hot-table scalar multiplication plus affine recovery; preparation and memory are reported separately.